RGB-D Salient Object Detection via Minimum Barrier Distance Transform and Saliency Fusion

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***Abstract*—Automatic detection of salient objects in images has gained its popularity in computer vision field for its usage in numerous vision tasks in recent years. Depth information plays an important role in the human vision system while it is underutilized in most existing two-dimensional (2-D) saliency detection methods. In this letter, a multistage salient object detection framework via minimum barrier distance transform and saliency fusion based on multilayer cellular automata (MCA) is proposed. First, we independently generate the 3-D spatial prior, depth bias, and RGB-produced and depth-induced saliency maps. Next, the two saliency maps are weighted by depth bias to obtain two initial maps. Then, we adopt a saliency optimization step to generate more precise depth-induced saliency map. Moreover, the initial RGB-produced and the optimized depth-induced maps are further fused with 3-D spatial prior. Finally, we utilize MCA to fuse all saliency maps generated previously and obtain the final saliency result with complete salient object. The proposed method is evaluated on the publicly available benchmark dataset, RGBD1000. Compared to several state-of-the-art 2-D and depth-aware approaches, the experimental results demonstrate the effectiveness and superiority of our method, which can accurately detect the salient objects from RGB-D images, and has the most satisfactory overall performance.**

***Index Terms*—Minimum barrier distance (MBD), RGB-D image, saliency fusion, saliency map, salient object detection.**

I. Introduction

VISUAL saliency detection is originally used to predict that where do people look at an image, and recently it was employed to highlight the conspicuous objects from the background in a scene, with the benefits for numerous applications including object detection and recognition [1], person reidentification [2], content-aware image resizing [3], image thumbnailing [4], etc. In this letter, we focus on object-level saliency detection. Several saliency maps generated by different methods are shown in Fig. 1.

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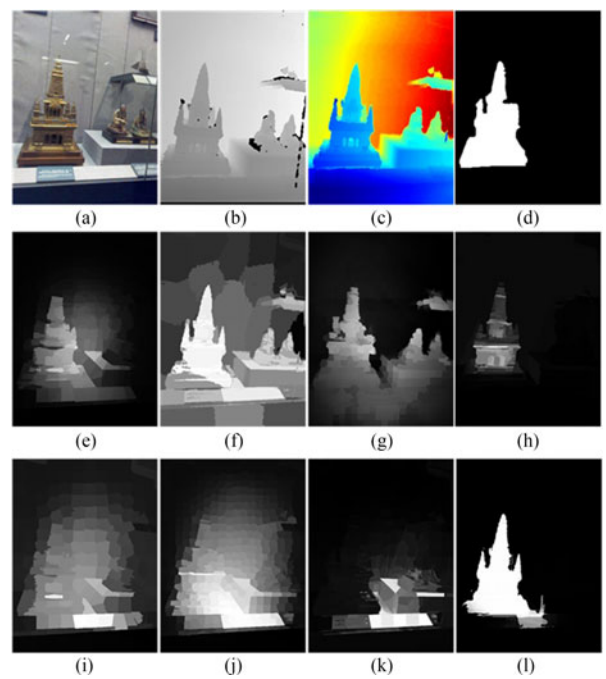

Fig. 1. Saliency maps generated by different methods. (a) Color image. (b) Depth map. (c) Depth map after preprocessing. (d) Ground truth. (e)–(k) Saliency maps generated by LMH [5], DES [6], ACSD [7], EGP [8], BSCA [9], GMR [10], and LPS [11], respectively. (l) Our result.

In 2007, Liu *et al.* [12] first propose a learning-based method with a set of novel features including multiscale contrast, center-surround histogram, and color spatial distribution to detect salient object in an image. Then, an enormous amount of excellent algorithms [13], [14] were proposed for salient object detection. Some two-dimensional (2-D) models compute local contrast [15], [16] or global contrast [17], [18] to identify salient object. Perazzi *et al.* [19] abstract image into homogeneous elements and estimate the uniqueness and the spatial distribution of these elements to obtain contrast-based saliency using high-dimensional Gaussian filters. Yang *et al.* [10] notice the phenomenon that background often presents local or global appearance connectivity with each of the four image boundaries. So, they proposed a graph-based manifold ranking method to formulate saliency of image superpixels. Li *et al.* [20] measure both dense and sparse reconstruction (DSR) errors of each image superpixels and propagate them based on the contexts obtained from clustering. Then, the results obtained are integrated by the Bayesian fusion method to generate the final saliency map. Zhang *et al.* [21] propose a fast minimum barrier distance (MBD) transform algorithm to measure boundary connectivity

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prior for fast salient object detection. This method runs at about 80 FPS using a single thread, and achieves comparable performance than several leading methods. In view of this, MBD is used as the base algorithm in our proposed framework. Qin *et al.* [9] introduce a novel propagation method based on cellular automata, denoted as BSCA. In BSCA, the single-layer cellular automata (SCA) can make an effective optimization toward the results of other 2-D methods via exploiting local similarity. In addition, the multilayer cellular automata (MCA) can take advantage of the superiority of other methods and incorporate multiple saliency maps to obtain a high-quality saliency map. Given the clear superiority of MCA, we use it as the selected fusion approach in our method. In brief, all of these methods focus on 2-D image. When confronted with challenging cases such as similar appearance between foreground and background, most of them possibly produce inaccurate saliency results.

As is known to all, human vision system (HVS) understands the real world from 3-D perception. Simultaneously, depth information has been proven to be helpful to many vision tasks including saliency detection, which provides not only the additional space layout information, but also the discriminative ability against the complex background. With the advent of the Kinect sensor [22], more and more depth-based saliency detection approaches [23]–[27] have been presented, which are devoted to utilize depth cue to boost performance. For instance, Niu *et al.* [23] extend global contrast method [18] for global disparity contrast analysis, and employ a photography knowledge that is specific to the stereopsis domain to identify salient object in stereo images. But, it will fail when a salient object has close oridentical depth with its surrounding. Lang *et al.* [24] extend existing 2-D methods by integrating depth that serves as the basis, and draw a conclusion that incorporating depth prior can significantly improve the quality of saliency maps. However, this method only leverages absolute depth value while local structure information is not taken into account, and evaluates the performance of eye fixation prediction models, rather than salient object detection methods. Desingh *et al.* [25] independently calculate region contrast on RGB images and depth maps, respectively, and then fuse them by nonlinear SVM regression to get the final saliency maps. Peng *et al.* [5] build a large-scale RGB-D dataset using Kinect, denoted as RGBD1000, and propose a multicontextual contrast model including local, global, and background contrast to compute depth saliency. They also propose a multistage RGB-D model, which is composed of low-level feature contrast, mid-level region grouping, and high-level prior enhancement. Ju *et al.* [7] propose a new depth feature, which is called anisotropic center-surround difference, to detect salient object from a single d p. Cheng *et al.* [6] combine the color contrast and depth contrast to detect salient object from RGB-D images. Meanwhile, 2-D center bias is extended to 3-D spatial bias, which is integrated with contrast saliency. Guo *et al.* propose a saliency fusion and propagation (SFP) strategy for RGB-D images, which fused multiple cue including color, location, and depth cue for high precision. Ren *et al.* [8] propose a two-stage RGB-D saliency detection framework, which first integrate multiple priors to generate an initial saliency map and then refine it via a restoration scheme to achieve the final result.

Still, the study leveraging depth cue for salient object detection has not been well exploited. Moreover, the priors induced by both 2-D image and 3-D information have not been well exploited when applied to RGB-D salient object detection in the existing models. Instead, these priors have shown their effectiveness in 2-D models such as [28], [10], and [29], by which comparable detection results on 2-D images are obtained. To this end, motivated by the aforementioned limitations and the superiority of these approaches discussed above, especially MBD and MCA, an effective multistage framework using multiple priors via MBD transform and MCA fusion strategy is proposed for RGB-D salient object detection. In this approach, background prior, depth bias, and 3-D spatial prior, which fused both location prior and depth cue, are taken into account and combined effectively. Moreover, distinguishing from the existing depth-aware methods, which simply combine several cues together, we integrate multiple saliency maps generated by various steps via MCA fusion strategy to obtain a final saliency map with better performance than several state-of-the-art methods. To sum up, the main contributions of our letter briefly lie in the following two aspects.

1) We proposed an effective multistage salient object detection framework, which includes the generation of initial saliency, optimization of saliency map, and multiple saliency fusion.
2) We effectively integrated multiple visual features and priors including background prior, 3-D spatial prior. and depth bias to enhance the performance of salient object detection remarkably.

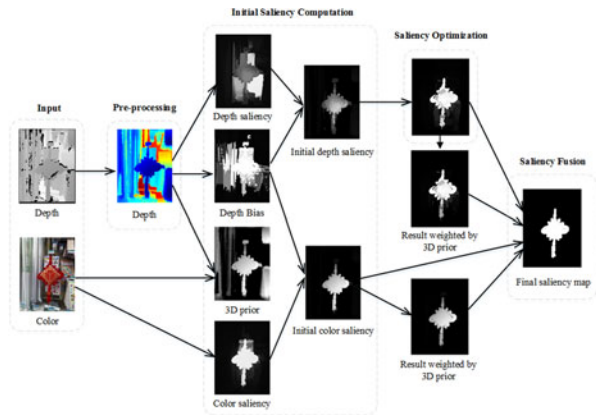

Fig. 2. Illustration of the main phases of our proposed algorithm for salient object detection in RGBD images.

II. Proposed Approach

Our approach mainly includes three parts: The first part conducts the estimation of initial RGB-based and depth-induced saliency. Next, several visual priors are computed, respectively, and incorporated with the initial saliency maps. Finally, we integrated these obtained maps to get the final saliency map. The proposed framework is shown in Fig. 2.

Considering that the original depth maps of RGB-D images is flawed, for example, holes with different sizes exist in a depth map commonly. It is quite necessary to repair the depth map for eliminating its impact on salient object detection methods. To this end, we adopt the rolling guidance filter, proposed in [30], to cope with this issue. And then, the restored results are further normalized to get smoother depth maps. Next, we compute both RGB-produced and depth-induced initial saliency maps via MBD, which has been proposed in [21]. For a 2-D single channel image $\mathcal{I}$, a path $\pi = \langle \pi(0), \ldots, \pi(k), \rangle$ is a sequence of image pixels where all of consecutive pixel pairs

are 4-adjacent. The path cost function of MBD is defined as follows:

$$\beta_I(\pi) = \max_{i=0}^{k} \mathcal{I}(\pi(i)) - \min_{i=0}^{k} \mathcal{I}(\pi(i)). \tag{1}$$

Zhang *et al.* [21] proposed an approximate iterative algorithm for the MBD transform, FastMBD, which is extremely efficient to generate a satisfying saliency map with linear complexity in the number of image pixels. So, we use it to measure the two initial saliency maps in this letter. Given the image I, the pixels along the image boundary is set as the background seeds *S*, a raster scan algorithm is applied to update distance map *D* until the required iterative number of passes is reached. Here the distance map *D* is the MBD path cost of some path that connects the seeds *S* and pixel *x*, defined in formula 1. For an RGB image, all MBD distance maps from each color channel are pixelwise added together to generate a saliency map S_c. Simultaneously, a depth map is a really single channel image, so we can directly use FastMBD method to calculate depth-induced saliency map S_d.

A. Computation of Visual Priors

Similar to the work by Guo *et al.* [31], depth bias is used to weight the two initial saliency maps obtained above, which is formulated as follows:

$$\omega_i^b = \begin{cases} 1, & d_i \geqslant t \\ \frac{d_i}{t}, & d_i < t \end{cases} \tag{2}$$

where d_i is the depth value of pixel p_i and t is the threshold for adjusting saliency values of p_i w.r.t. its depth value, which is set as the median depth value of each depth map. After the depth bias ω_i^b is calculated, we further compute two weighted saliency maps S_c^d and S_d^d by multiplying the two initial saliency maps derived before by depth bias. In addition, location prior, also known as center bias, which has been extensively adopted to boost performance of salient object detection methods. In DES [6], center bias is extended into 3-D spatial prior and is calculated as

$$\omega_i^{3-D} = \frac{1}{N}\mathcal{G}\left(\| \mathrm{Loc}_i - C\|_2 | 0, \sigma^2\right) \cdot (\max d_i - d_i)^{\gamma/\mathrm{DOF}} \tag{3}$$

where N, C, and σ^2 are the pixel number, center, and normalized radius of each RGB image I, respectively, $\mathcal{G}(\cdot)$ is the Gaussian kernel, which is used to assign a relatively larger saliency values to these pixels closer to the image center C, Loc_i represents the coordinates of pixel p_i, DOF, namely depth-of-field, is the difference between the maximum and minimum depth value of the corresponding depth map.

B. Saliency Optimization

In fact, the depth-induced saliency results weighted by depth bias are still undesirable. Naturally, a saliency optimization step is conducted to improve these undesirable saliency maps in this letter. For this purpose, we leverage the SCA method [9] to cope with this problem. But, it is noted that the SCA method is designed for RGB-based saliency map. Considering the difference between RGB-based map and depth-induced map, we made a modification to the SCA method as follows. The original impact factor matrix F is redefined as

$$f_{ij} = \begin{cases} \exp\left(\dfrac{-\|d_i, d_j\|}{\sigma_f^2}\right), & j \in NB(i) \\ 0, & i = j \text{ or otherwise} \end{cases} \tag{4}$$

where $\|d_i, d_j\|$ is the absolute difference of between depth values d_i and d_j, which means that neighbors with more similar depth values have a greater influence on current cells next state. Same as [9], σ_f^2 is a parameter to control strength of similarity and is also set to 0.1. Through this optimization step, a better depth-induced saliency map is obtained, denoted as S_d^{do}. The next step is to calculate the combined saliency maps S_c^{3-D} and S_d^{3-D} by multiplying the two saliency maps S_c^d and S_d^{do} by 3-D spatial prior.

C. Multiple Saliency Fusion

In order to make full use of the respective characteristics of each saliency map generated in various stages, we incorporate these saliency maps by the aid of MCA [9]. As a more effective saliency fusion method, MCA produces more satisfactory fusion results than the linear combination strategy adopted by many previous saliency detection algorithms. In MCA, a pixel is represented as a cell, pixels with the same coordinates in different saliency maps are neighbors. The saliency value of cell c_i is treated as its probability to be the foreground F, denoted as $P(c_i \in F) = S_i$ while $P(c_i \in B) = 1S_i$ stands for its possibility to be the background B. An adaptive threshold segmentation method based on OTSU [32] is used to binarize each saliency map, and η_i represents the state of cell c_i

$$\eta_i = \begin{cases} +1, & S_i \geqslant \gamma^j \\ -1, & S_i < \gamma^j \end{cases} \tag{5}$$

where γ^j is the threshold of the *j*th saliency map. Next, the posterior ratio $\Lambda(c_i \in F|\eta_j = +1)$ is defined as

$$\Lambda(c_i \in F|\eta_j = +1) = \frac{S_i}{1 - S_i} \cdot \frac{\lambda}{1 - \mu} \tag{6}$$

where $\lambda = P(\eta_j = +1|c_i \in F)$ and $\mu = P(\eta_j = -1|c_i \in B)$ are the probabilities of a neighbor c_j to be F and B respectively when cell c_i belongs to F. Accordingly, the synchronous updating rule of saliency fusion is defined as

$$\ln(\mathbf{S}_m^{t+1}) = \ln(\mathbf{S}_m^t) + \sum_{k=1, k\neq m}^{M} \mathrm{sign}(\mathbf{S}_k^t - \gamma^k \cdot \mathbf{L}) \cdot \ln\left(\frac{\lambda}{1-\lambda}\right) \tag{7}$$

where $\mathbf{S}_m^t$ represents the saliency value of all cells on the *i*th saliency map at time *t*, and the matrix $\mathbf{L} = [1, 1, \ldots, 1]^T$ have H elements. H is the pixel number of an image. After N time steps, the final fused saliency map $\mathbf{S}^*$ is formulated as

$$\mathbf{S}^* = \frac{1}{M}\sum_{m=1}^{M} \mathbf{S}_m^N. \tag{8}$$

In this letter, we incorporate five saliency maps generated from different stages to obtain the final saliency map. It includes S_c, S_c^d , S_d^d, S_c^{3-D}, and S_d^{3-D}.

TABLE I
PRECISION, RECALL, F-MEASURE, MAE, AND WFB VALUES OF MAIN COMPONENTS IN THE PROPOSED METHOD

Component	S_c	S_c^d	S_c^{3-D}	S_d	S_d^d	S_d^{do}	S_d^{3-D}
Precision	0.606	0.667	0.701	0.475	0.588	0.666	0.694
Recall	0.786	0.733	0.697	0.567	0.652	0.708	0.717
F-measure	0.613	0.652	0.671	0.466	0.568	0.645	0.669
MAE	0.154	0.106	0.095	0.183	0.124	0.144	0.098
WFb	0.410	0.467	0.450	0.284	0.363	0.443	0.526

III. EXPERIMENTAL RESULTS AND ANALYSIS

To illustrate the effectiveness of the proposed approach, we conduct qualitative and quantitative comparison and analysis between the proposed algorithms and the state of the art. The authoritative precision-recall curve, F-measure and mean absolute error (MAE) [19] are used to evaluate our method on RGBD1000 dataset. In addition to the three metrics, a more reasonable evaluation approach, namely, weighted F-measure [33], denoted as WFb, is calculated in this letter.

A. Saliency Results and Discussion of the Proposed Method

The effectiveness of each component in our approach is first evaluated. As shown in Table I, with the integration of depth bias and 3-D spatial prior successively, the precision, F-measure, MAE, and WFb values are increased consistently w.r.t. both RGB-based and depth-induced saliency, except that the recall is decreased w.r.t. RGB-based saliency. It is probably because multiplication operation is not the best way to combine these priors with the RGB-based saliency. In a word, depth bias and 3-D spatial prior can significantly enhance the performance of saliency maps in our fusion framework. In addition, the effect of saliency optimization is also verified in Table I. All the five evaluation metrics including precision, recall, F-measure, MAE, and WFb scores are improved consistently for the depth-induced saliency. So, it is worthy to conduct this optimization step for salient object detection.

B. Comparison With the State-of-the-Art Methods

We further make a comprehensive comparison of our algorithm with the thirteen 2-D and five 3-D methods. We show a few examples of saliency maps generated by different methods in Fig. 3. The first row of Fig. 3 shows that all of the methods can accurately detect the salient object in a relative ideal image. The second row shows that a few depth-induced methods such as ACSD [7] and SFP [34] are invalid. It may be because ACSD only takes depth cue into consideration. As for SFP, it is likely that the elementwise dot product is utilized for saliency fusion. In the third and fourth rows, most RGB-based models such as MBD + [21] failed to identify the conspicuous object. This is mainly because RGB-based models ignore the helpful depth cue. In the fifth and sixth rows, many 2-D and 3-D methods experienced poor performance in some challenging cases. This explains that effectively combining the visual features is very important for saliency analysis. Instead, our method can correctly detect the salient object in all of these examples by leveraging multiple visual features appropriately.

Next, we carry out a qualitative analysis of our method and these excellent models. The resulting precision, recall and F-measure values of the proposed approach are superior to others. Meanwhile, our method also obtain the lowest MAE value and the highest WFb score. Due to space constraints, we only give the MAE and WFb values in Fig. 4(a)–(d).

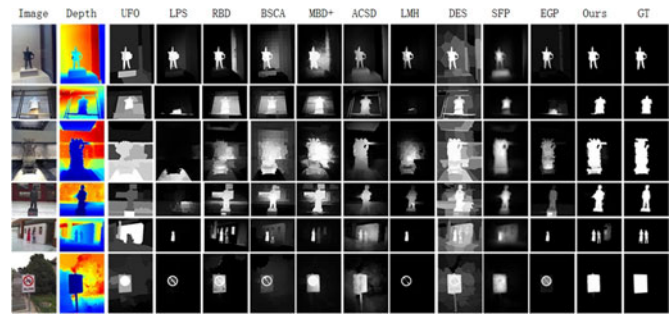

Fig. 3. Examples of comparison with the state-of-the-art 2-D and depth-aware methods. From left to right: color image, depth map, saliency maps generated by UFO [35], LPS [11], RBD [29], BSCA [9], MBD + [21], ACSD [7], LMH [5], DES [6], SFP [34], EGP [8], and ours, respectively, GT.

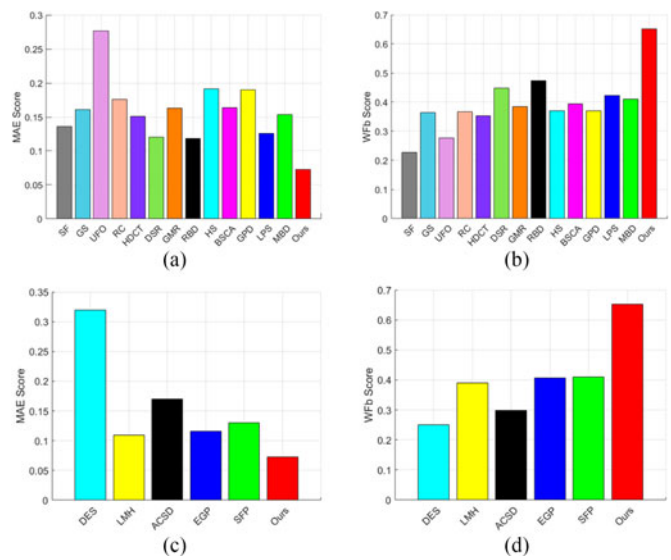

Fig. 4. Performance comparisons of the proposed algorithm versus thirteen 2-D saliency models including SF [19], GS [28], UFO [35], RC [18], HDCT [36], DSR [20], GMR [10], RBD [29], HS [16], BSCA [9], GPD [37], LPS [11], and MBD + [21], and five depth-induced models including DES [6], LMH [5], ACSD [7], EGP [8], and SFP [34]. (a) MAE scores of 2-D models. (b) WFb scores of 2-D models. (c) MAE scores of 3-D models. (d) MAE scores of 3-D models.

IV. CONCLUSION

In this letter, we proposed a multistage fusion framework integrating multiple visual priors from RGB image and depth map for salient object detection. Specifically, we deeply investigated how to effectively combine these useful visual features for a higher quality of saliency map. Moreover, we adopted a variety of evaluation metrics to conduct a comprehensive assessment of the state of the art and the proposed method. The experimental results demonstrated that the proposed method achieves superior performance and more satisfying results than other competing 2-D and 3-D methods.

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